

BA820 – Project Milestone 2

Section A1, Team 7

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Uncovering Structural Patterns in High-Stakes Regulatory and Funding Decisions for Drugs
(European Drug Development)

1. Project Motivation Shifts Motivated From Initial EDA:

1.1 Project Motivation, What Changed:

The development and regulatory evaluation of pharmaceutical products is a costly and uncertain process, particularly at the late-stage point where drug candidates are submitted for regulatory approval. At this stage, firms must allocate limited resources across competing candidates while regulators balance timely patient access with evidentiary standards for safety and efficacy.

The European Medicines Agency (EMA) oversees centralized [marketing authorization](#) for medicines in the European Union and offers alternative approval pathways alongside standard review, including accelerated assessment, conditional marketing authorization, and orphan designation for rare diseases. These mechanisms are intended to encourage innovation and address unmet medical needs, but they also involve trade-offs related to evidentiary uncertainty and regulatory risk.

This particular section of the project's focus will be a narrowed-down focus on understanding the semantic landscape of drug indications. Official therapeutic_area labels are revealed to be inconsistent for meaningful segmentation between drugs, leading to further examinations such as condition_indication that indicate use cases and conditions on a much more granular level.

1.2 Specific Focus: Cross-Domain Exploration Between Human and Veterinary Medicines

Further investigation into the specific focus aims to examine and evaluate opportunities for cross-domain drug production between human and veterinary medicines. Initial EDA revealed condition_indication to have extensive text data that shows potential for overlap in regulatory phrasing, allowing medicines to be grouped by shared therapeutic intent. Furthermore, examining product portfolio categories of the largest players in the European pharmaceutical market reveals separate firms specializing in either human or veterinary medicines.

While liability profiles and regulatory requirements differ for human and animal drugs, leading to specialization in only one domain for many firms, parent companies of such firms remain an overarching entity, revealing opportunities for portfolio optimization by reducing R&D time and early-stage testing costs. “Blue Ocean” opportunities (areas with high innovation but low competition) may also exist between these domains, which can be capitalized upon for portfolio expansion. Standard high level categories can hide these nuances; for example, a “Diabetes” drug could be a generic pill or a novel peptide that aims to solve its root cause. By using Unstructured Machine Learning (NLP) on the condition_indication text, data-driven patient segments that reveal the true structure of the pharmaceutical market can be discovered.

Methodology & Approach:

Data Preprocessing:

- **Symbol Preservation:** Medical text relies on symbols (e.g., *HER2+*, *CD30*, Stage IV). Standard cleaners remove special symbols such as these, which risks losing out on critical data. I implemented a custom pipeline replacing symbols with text (e.g., *+* → plus) to preserve biomarkers like *cd30plus* as distinct features.
- **Lemmatization:** Words like "treating" and "treatment" were unified to their root to reduce dimensionality.
- **TF-IDF Vectorization:** I generated a matrix of the top 1,000 significant words, filtering out common "stop words" (e.g., *patient*, *indicated*) that appear in >95% of documents to focus purely on differentiating medical terms.

Unsupervised Techniques:

- **K-Means Clustering:** Used to segment drugs into *K* distinct semantic groups. The **Elbow Method** (via Yellowbrick) was used to determine the optimal *K* value of **16**.
- **t-SNE (Dimensionality Reduction):** Used to visualize the 1,000-dimensional TF-IDF matrix in 2D space. `init='pca'` was used to preserve global structure better than random initialization.

Findings, Insights, and Interpretations:

Finding A: Hidden Clusters

- The K-Means algorithm successfully identified distinct therapeutic clusters that align with biology and patient type, cutting across official administrative categories
 - **Cluster [1] (Cardiovascular Cluster):** Defined by keywords [*hypertension*, *blood*, *pulmonary*, *arterial*]. This cluster isolates cardiovascular treatments, specifically for blood pressure regulation. The model successfully separated these hemodynamic agents from other metabolic drugs such as “diabetes”.
 - **Cluster [15] (Veterinary Cluster):** Defined by keywords [*treatment*, *adult*, *dog*, *severe*, *chronic*]. The keyword “dog” is a very important point of interest. Combined with other words such as *chronic*, this finding demonstrates that the model is able to distinguish between human and animal purely based on text data.

Finding B: Blue Ocean Opportunities

- The t-SNE visualization revealed several “semantic islands”-- drugs that float far away from main clusters.

- **Observation:** Top outliers were consistently related to bleeding and *haemophilia*, which seemed to have formed their own cluster that are removed from “cardiovascular” or “oncology” clusters
- **Interpretation:** Further research into haemophilia treatments revealed a highly specialized treatment plan, and often uses unique vocabulary that shares little linguistic overlap with standard cardiovascular complication drugs. These represent a specific niche that the model has isolated.

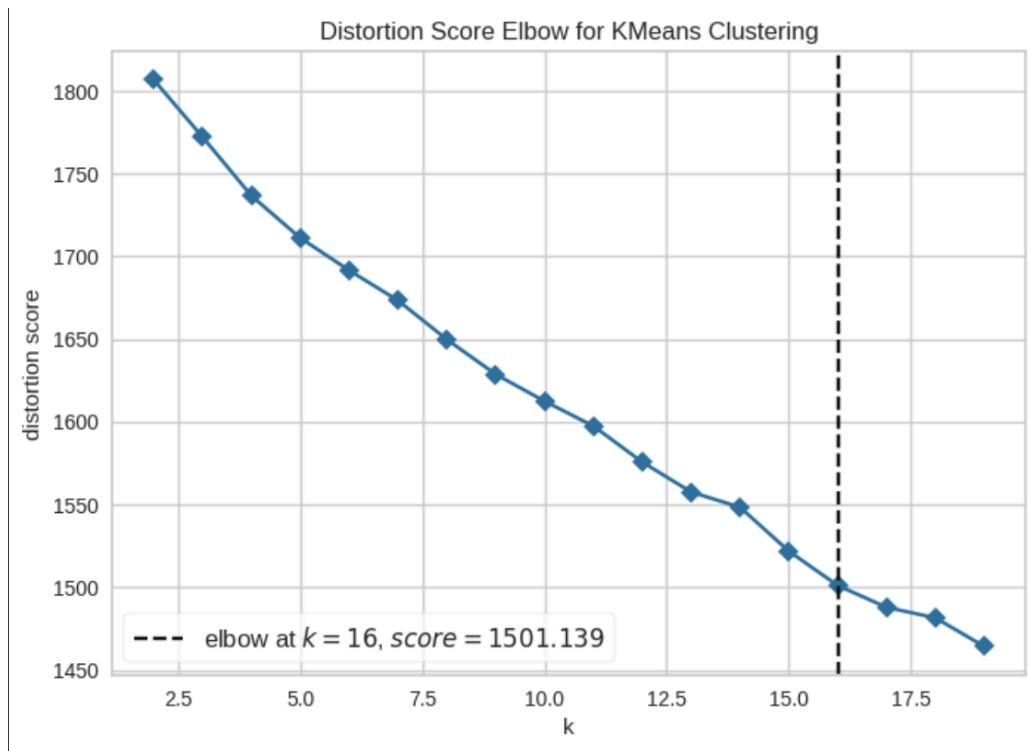
Finding C: Corporate Silos

- Cross tabular examination of the K-Means clusters against official administrative categories (humans vs. animals)
 - **Insight:** The heatmap revealed proof of distinct separation between human and veterinary that the model was able to identify, as well as opportunities for cross-domain use cases. Cluster 15 contains a mix of 419 human drugs to 146 veterinary drugs, whereas Cluster 13 shows a much more distinct split with 7 human drugs and 131 veterinary drugs.
 - **Interpretation:** The model gives proof of concept in the distinctions between describing animal and human health, as well as opportunities for drug repurposing. Corporate silos may exist in the legal sense, but semantic silos are not as clear for human and dog chronic pain as the mechanism is virtually identical.

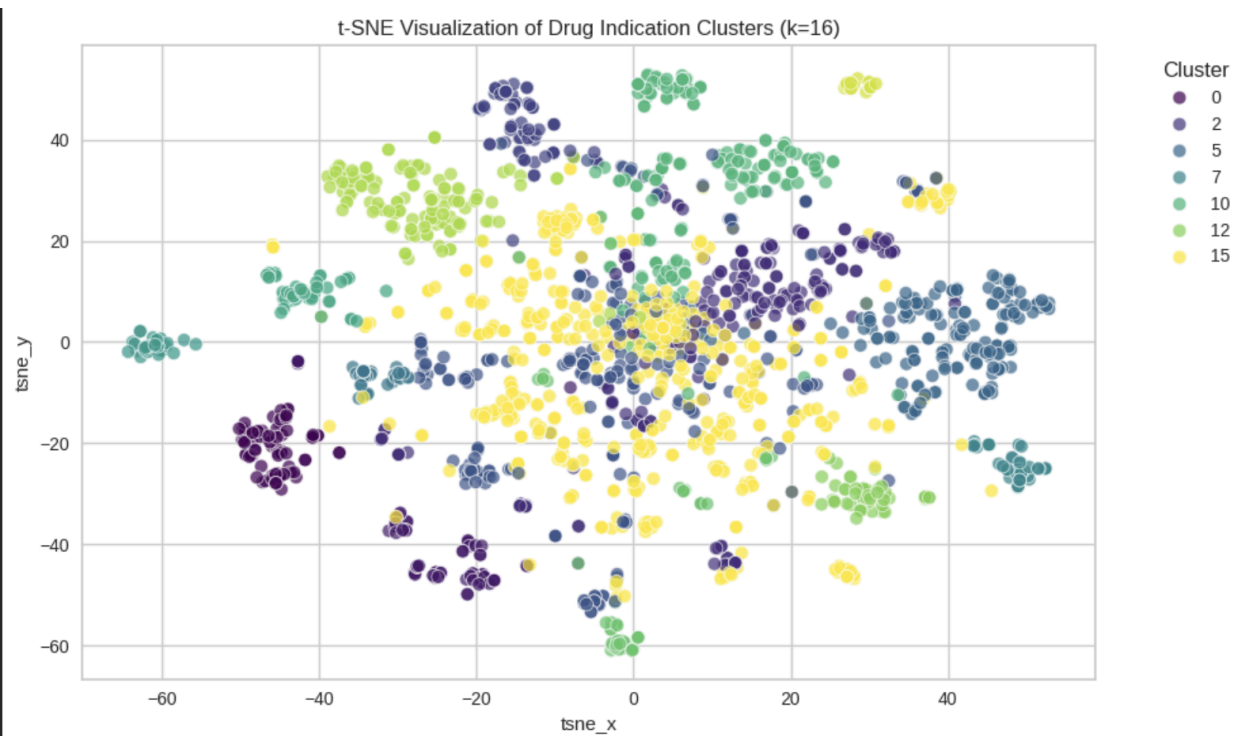
Next Steps:

1. **Refine NLP:** I will refine the NLP model to understand the bridges between “canine lymphoma” and “human lymphoma” as similar, even as patient species differs.
2. **Anomaly Detection:** I will look to apply an Isolation Forest method to apply a “novelty score” of each drug. This step will aim to convert the outliers into a ranked list that can offer the highest probability of identifying a “Blue Ocean”.

Appendix:



Optimized K-Clusters



t-SNE Plot With 16 Identified Clusters

Cluster 0:
metformin, control, glycaemic, diet, exercise, mellitus, diabete, combination, type, adjunct, patient, adequate, diabetes, insulin, sulphonylurea, provide

Cluster 1:
hypertension, essential, pressure, adequately, blood, patient, control, treatment, pulmonary, adult, arterial, disease, indicate, class, combination, mg

Cluster 2:
patient, adult, treatment, indicate, therapy, leukaemia, combination, refractory, lymphoma, prior, cell, monotherapy, disease, receive, myeloma, chronic

Cluster 3:
official, accordance, influenza, recommendation, vaccine, cause, age, year, immunisation, use, individual, active, prophylaxis, the, virus, guidance

Cluster 4:
year, age, treatment, adolescent, indicate, old, patient, adult, child, deficiency, 12, seizure, therapy, paediatric, syndrome, epilepsy

Cluster 5:
cancer, metastatic, advanced, treatment, patient, cell, indicate, locally, combination, chemotherapy, breast, lung, carcinoma, adult, monotherapy, therapy

Cluster 6:
mellitus, diabete, child, treatment, adolescent, year, age, adult, insulin, require, indicate, diabetes, initial, maintenance, severe, achieve

Cluster 7:
fracture, osteoporosis, risk, bone, woman, postmenopausal, vertebral, increase, hip, man, treatment, reduction, event, surgery, prevention, significant

Cluster 8:
haemophilia, congenital, bleed, deficiency, prophylaxis, factor, patient, group, treatment, age, contain, episode, 12, severe, indicate, year

Cluster 9:
infarction, myocardial, day, coronary, asa, atherothrombotic, event, acute, prevention, clopidogrel, patient, stroke, suffer, elevation, segment, acid

Cluster 10:
infection, use, antiretroviral, agent, product, treatment, medicinal, hivminus1, patient, adult, indicate, combination, antibacterial, appropriate, hepatitis, give

Cluster 11:
symptom, obstructive, copd, pulmonary, maintenance, chronic, adult, treatment, indicate, disease, relief, patient, acid, associate, severe, moderate

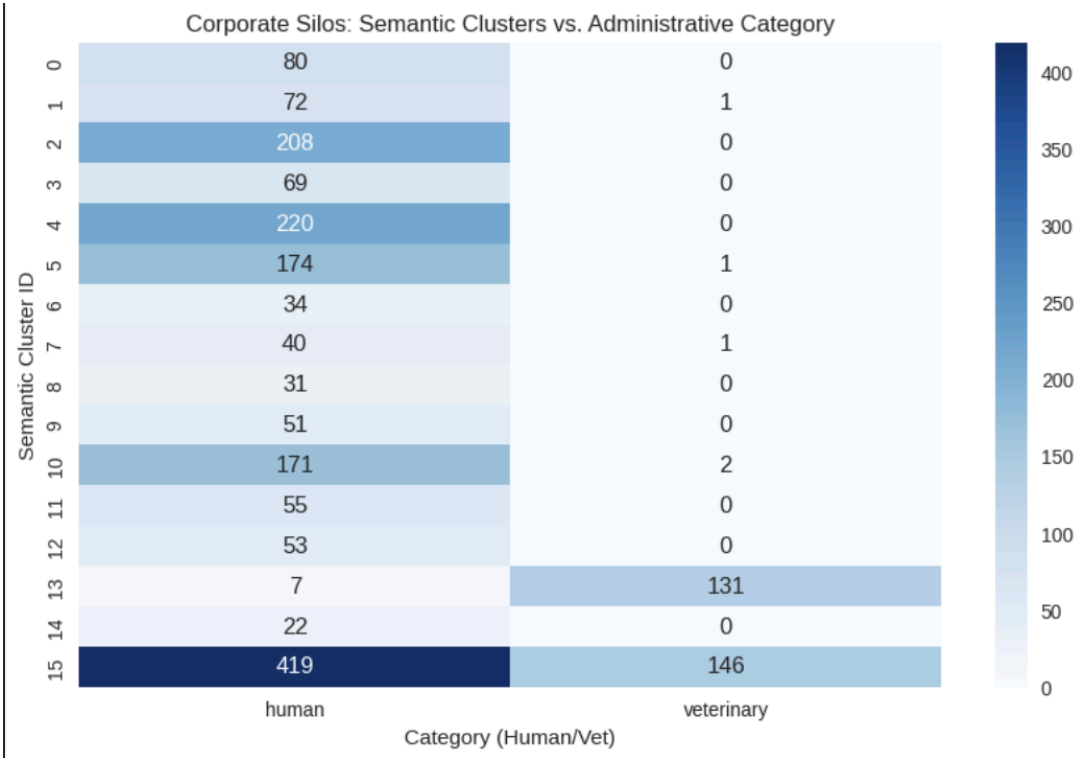
Cluster 12:
arthritis, methotrexate, inadequate, response, active, intolerant, rheumatoid, therapy, treatment, dmard, psoriatic, conventional, severe, drug, adult, patient

Cluster 13:
immunisation, reduce, active, clinical, cause, sign, virus, immunity, week, mortality, infection, prevent, vaccination, age, lesion, month

Cluster 14:
neutropenia, duration, incidence, cytotoxic, myelodysplastic, malignancy, myeloid, chemotherapy, reduction, leukaemia, syndrome, treat, chronic, patient, count, infection

Cluster 15:
treatment, adult, indicate, patient, pain, symptomatic, disease, associate, severe, dog, prevention, moderate, disorder, acute, chronic, therapy

Cluster Keywords



Cluster Segmentation Against Higher Level Categories

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--- Insight B: The 'Blue Ocean' Outliers ---
Drug: Obizur
Condition: Treatment of bleeding episodes in patients with acquired haemophilia caused by antibodies to Factor ...
Cluster: 8
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Drug: NovoSeven
Condition: NovoSeven is indicated for the treatment of bleeding episodes and for the prevention of bleeding in ...
Cluster: 8
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Drug: Cevenfacta
Condition: CEVENFACTA is indicated in adults and adolescents (12 years of age and older) for the treatment of b...
Cluster: 8
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Drug: HemLibra
Condition: Hemlibra is indicated for routine prophylaxis of bleeding episodes in patients with haemophilia A (c...
Cluster: 8
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Drug: Roctavian
Condition: Treatment of severe haemophilia A (congenital factor VIII deficiency) in adult patients without a hi...
Cluster: 8
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Drug: Hemgenix
Condition: Treatment of severe and moderately severe Haemophilia B (congenital Factor IX deficiency) in adult p...
Cluster: 8
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Drug: Coagadex
Condition: Coagadex is indicated for the treatment and prophylaxis of bleeding episodes and for perioperative m...
Cluster: 8
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Drug: NovoEight
Condition: Treatment and prophylaxis of bleeding in patients with haemophilia A (congenital factor VIII deficie...
Cluster: 8
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Drug: Iblias
Condition: Treatment and prophylaxis of bleeding in patients with haemophilia A (congenital factor VIII deficie...
Cluster: 8
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Drug: Kovaltry
Condition: Treatment and prophylaxis of bleeding in patients with haemophilia A (congenital factor VIII deficie...
Cluster: 8

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Outlier Cluster Detection

AI Disclosure Statement:

In this section of the project, Gemini 3 model was used in order to map out workflows according to the rubric, as well as implementation of the t-SNE model, as it was mentioned in class, but wasn't expanded upon in terms of implementation. I attempted to engage with Gemini's output, correcting its readings and interpretations in order to fit my notebook, as well as asking follow up questions on its methodologies and code that I failed to understand on initial readings.

Chat Link: <https://gemini.google.com/app/92f2c2fed98ad8fe>